

Generative AI Prompt Engineering – Hands-On Exercises

Below are **3 BMW-oriented exercises for each prompt-engineering category**, with a hint and suggested tools.

Category	Exercise	Hands-On Task	Hint	Tools
Zero-Shot Prompting	1. Vehicle Health Classification	Give battery level, temperature and fault code. Ask AI to classify Normal / Warning / Critical without examples.	Define the task and required output clearly; do not give sample classifications.	ChatGPT Playgrou Studio
	2. Service Ticket Classification	Give a customer complaint such as “My BMW iX battery drains overnight.” Ask AI to classify it as Battery / Software / Charging / Mechanical.	Specify the allowed categories.	ChatGPT Microsof
	3. Manufacturing Defect Analysis	Provide a short production-line defect description and ask AI to identify the defect category and severity.	Include input + task + output format, but no examples.	ChatGPT Jupyter
Few-Shot Prompting	1. Vehicle Fault Classification	Give 3 examples of Normal, Warning and Critical telemetry, then ask AI to classify a fourth vehicle.	Examples should closely resemble the target problem.	ChatGPT Playgrou
	2. Customer Query Routing	Give examples such as “Charging issue → EV Support” and “Navigation issue → Infotainment Support,” then classify a new complaint.	Use consistent example formatting.	ChatGPT Studio
	3. Quality Defect Classification	Give examples of Paint Defect, Welding Defect and Assembly Defect. Ask AI to classify a new manufacturing incident.	Include both input and expected label in every example.	ChatGPT
Structured Reasoning / Chain-of-Thought Style	1. Battery Diagnosis	Give low battery, high temperature and an overheating fault. Ask AI to identify the relevant factors and provide a brief diagnostic rationale + conclusion .	Ask for decision factors and a concise rationale rather than hidden internal reasoning.	ChatGPT

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	2. Vehicle Range Investigation	Vehicle range suddenly decreases. Ask AI to consider battery health, weather, driving style and HVAC usage before giving a likely cause.	Ask for: Factors → Evidence → Conclusion.	ChatGPT Studio
	3. Manufacturing Downtime Analysis	Provide machine alarms, cycle time and maintenance history. Ask AI for a structured root-cause assessment.	Make the reasoning observable using a checklist or table.	ChatGPT Foundry
	1. Battery Overheating	Ask AI to explore three hypotheses: cooling-system problem, sensor problem and high-load driving. Compare them and choose the best-supported explanation.	Use Hypothesis → Evidence → Confidence → Decision .	ChatGPT
Tree of Thoughts	2. Production Defect Investigation	Explore whether a defect comes from machine calibration, raw material or operator process.	Ask AI to rank alternative branches rather than immediately choosing one.	ChatGPT
	3. Customer Complaint Diagnosis	For “vehicle will not charge,” consider charger issue, cable issue, battery issue and software issue.	Have each branch evaluated against supplied evidence.	ChatGPT
Prompt Chaining	1. Customer Complaint Resolution	Prompt 1: Identify intent → Prompt 2: Extract facts → Prompt 3: Analyze cause → Prompt 4: Recommend action → Prompt 5: Format response.	Output of each prompt should become input to the next.	ChatGPT LangGra
	2. Telemetry Report Generation	Extract anomalies → classify severity → identify likely causes → generate service recommendations → create final report.	Keep each prompt responsible for one task.	LangCha Jupyter
	3. BMW Sales Analysis	Extract sales data → calculate trends → identify top models → explain changes → produce executive summary.	Validate intermediate results before passing them forward.	ChatGPT Pandas

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Self-Consistency	1. Fault Diagnosis Comparison	Generate several independent diagnoses for the same telemetry, then select the conclusion supported by the strongest common evidence.	Compare final conclusions, not hidden reasoning traces.	ChatGPT
	2. Warranty Classification	Ask the same warranty case several times with controlled variation and compare the classifications.	Use majority agreement or an evaluation rule.	OpenAI /
	3. Root-Cause Validation	Generate three candidate root causes for production downtime and ask a separate evaluator prompt to rank them.	Separate generation from evaluation .	LangCha Python
Role Prompting	1. Diagnostic Expert	Start with: “You are a BMW vehicle diagnostic expert.” Ask it to analyze telemetry.	Define expertise, audience and expected responsibility.	ChatGPT Playgrou
	2. BMW Data Analyst	“You are an automotive data analyst.” Ask for trends from vehicle sales or telemetry data.	Specify business audience: technician, manager or customer.	ChatGPT
	3. Service Advisor	“You are a BMW service advisor.” Convert technical fault details into a customer-friendly explanation.	Control both role and tone.	ChatGPT
Instruction Following	1. Strict Output Format	Ask AI to return only <code>Vehicle ID Severity Action</code> .	Add “Do not include additional explanation.”	ChatGPT Playgrou
	2. Data Restrictions	Tell AI to analyze only battery level and temperature and ignore all unrelated fields.	Clearly state allowed and prohibited behavior.	ChatGPT Foundry
	3. JSON Response	Ask AI to return vehicle diagnostics as valid JSON with specific keys.	Define the schema explicitly.	OpenAI f Python, f

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Avoiding Hallucinations	1. Service Manual Q&A	Provide a short BMW maintenance document and ask questions that are partly answered and partly absent.	Add: “If not present, respond Insufficient information.”	ChatGPT RAG
	2. Unknown Fault Code	Give an invented fault code such as BMW_XYZ_999 . Test whether AI admits that evidence is insufficient.	Explicitly prohibit inventing fault-code definitions.	ChatGPT
	3. Grounded Vehicle Recommendation	Supply an approved list of vehicle specifications and ask AI to compare models using only that list.	Require evidence from provided context for every recommendation.	ChatGPT LangSmi
Iterative Prompt Refinement	1. Improve a Weak Diagnostic Prompt	Start with “Analyze this BMW.” Improve it through 3 versions.	Add role → context → task → constraints → format.	ChatGPT
	2. Reduce Verbosity	Start with a prompt producing a long service report. Refine it until output fits within 5 lines.	Add length, format and audience constraints.	ChatGPT Playgrou
	3. Improve Classification Accuracy	Start with an ambiguous fault-classification prompt. Test it, identify errors, refine wording/examples and test again.	Keep versions V1, V2 and V3 and compare outputs.	LangSmi ChatGPT

Suggested Lab Pattern

Use the same process for every exercise:

1. Write Prompt → 2. Run Prompt → 3. Observe Output → 4. Identify Problems → 5. Improve Prompt → 6. Compare Results

For BMW training, participants can record results in a simple table:

Prompt Version	Technique	Result	Problem Found	Improvement
V1	Zero-shot	Warning	Output too verbose	Add fixed format
V2	Instruction	Warning	Cause invented	Restrict to supplied data
V3	Refined	Warning	Acceptable	Final version

Recommended Core Toolset for the Labs

Beginner: ChatGPT, Claude, Google AI Studio

LLM experimentation: OpenAI Playground, Microsoft Foundry

Prompt management/evaluation: LangSmith, Promptfoo, Langfuse

Developer labs: Python/Jupyter, VS Code, LangChain, LangGraph

RAG labs: LangChain + FAISS/Chroma + an LLM API

Important for the Chain-of-Thought module: teach participants to request **concise rationale, decision factors, hypotheses, evidence, or structured intermediate outputs** rather than asking a production model to expose private internal reasoning. This achieves the educational goal while producing outputs that can actually be evaluated.